

ICI

- Resistors (all
- $\frac{1}{4}$
- watt,
- ± 5
- per cent carbon):

- Capacitors:

- Miscellaneous:*

- | | |
|-------------------|----------------------------------|
| X _{TAL1} | - 11.0592MHz crystal |
| SI | - Push-to-on switch |
| PZ1 | - Piezobuzzer |
| Board | - EFY-KrS 8051 development board |

The water level in the tank is sensed by the sensor assembly, which is connected to 4-bit data lines (AD8 through AD11) of encoder HT12E (ICI) through transistors. Depending on the water level in the tank, BC547 transistors (T1 through T4) conduct to generate a 4-bit code (refer the table). The 4-bit code so generated is encoded by encoder

4-bit Code, Message and Buzzer Status for Different Water Levels

Water level	4-bit code (DCBA)	Message displayed on the LCD	Buzzer status
Less than 25 per cent	0000	Water level low	Buzzer rings for two minutes
25 per cent	1000	Water level 25 per cent	—
50 per cent	1100	Water level 50 per cent	—
75 per cent	1110	Water level 75 per cent	—
100 per cent	1111	Water level 100 per cent	Buzzer rings for two minutes

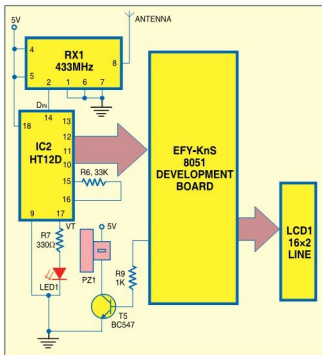


Fig. 3: Testing of receiver circuit using development board

The 4-bit signal from the transmitter is received by the antenna of the RF receiver (RX1). LED1 glows to indicate that a valid signal is received. The 4-bit output from decoder HT12D is processed by the microcontroller to generate an 8-bit code. The microcontroller's output is fed to data input lines of LCD1, which, in turn, shows the water level in percentage.

At an intermediate water level, say, 25 per cent of the tank capacity, LCD1 shows the message "Water level

25 per cent. When the tank is full, the buzzer rings for two minutes, while LCD1 shows "Water level 100 per cent. The 4-bit code, message and buzzer status for different water levels are shown in the table.

Software

In the program, first LCD1 and buzzer are initialised followed by the reset, read-write and enable pins of LCD1. Then LCD1 shows 'EFY' in the first line and 'Water

Level' in the second line. The program further enters a loop to check which of the five values is true—less than 25 per cent, 25 per cent, 50 per cent, 75 per cent or 100 per cent. The percentage is displayed in the second line after 'Water Level.' The piezobuzzer is timed to sound for two minutes for the 100 per cent full condition and less than 25 per cent full condition.

This circuit can also be modified to work as a water-level controller system. When the water level is low, the microcontroller can be programmed to start the motor pump. When the tank is full, the same can be made to stop the motor.

EFY note. The source code of this article is available on www.efymag.com website. ●